Birthdays, Derangements, and a Little Calculus

50 minutes • 50 points • Student quiz

Name: _____ **Date:** _____

Instructions. Show your reasoning. Give exact fractions where a number is requested; no calculator is needed. You may use $\ln 2 \approx 0.693$ and $1 + 2 + \cdots + (n-1) = n(n-1)/2$ without proof. The suggested times leave 5 minutes of slack.

1. The calculus toolkit

10 points · about 8 minutes

(a) Write the Taylor series of e^x about $x = 0$: give the terms through x^3 and the general term. Then write $1/e$ as an infinite series of numbers. (3 points)

(b) Let $f(x) = \ln(1 - x)$. Find $f(0)$ and $f'(0)$, and write the tangent-line approximation to $f(x)$ for x near 0. (3 points)

(c) Find $\lim_{n\to\infty} n \ln\left(1 - \frac{1}{n}\right)$, and use it to find $\lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^n$. Justify the first limit (the definition of $f'(0)$ from part (b), or L'Hôpital's rule, will do). (4 points)

2. Shared birthdays on a d-day calendar

14 points · about 13 minutes

A year has d days. People enter a room one at a time; their birthdays are independent and uniformly distributed over the d days. Let N_n be the event that the first n people all have *different* birthdays.

(a) Suppose the first k people have k different birthdays. What is the conditional probability $\mathbb{P}(N_{k+1} \mid N_k)$ that person $k+1$ misses all of them? Use the multiplication rule to write $\mathbb{P}(N_n)$ as a product (for $n \le d$). (4 points)

(b) Take the logarithm of your product and use $\ln(1-x) \approx -x$ to show that, when n is much smaller than d, (4 points)

$$\mathbb{P}(N_n) \approx \exp\Big(-\frac{n(n-1)}{2d}\Big).$$

(c) On a planet whose year has $d = 100$ days, use the approximation in (b) to find the smallest n for which a shared birthday is predicted to be more likely than not. (3 points)

(d) Now only three people are in the room. Find the probability that all three share one birthday, *given* that at least two of them do. Simplify your answer as a function of d. (3 points)

3. The hat-check problem

16 points · about 14 minutes

Each of n guests checks a hat, and the hats are returned in a uniformly random order: all $n!$ assignments are equally likely. Let A_i be the event that guest i receives their own hat, and let p_n be the probability that *nobody* receives their own hat (the assignment is a *derangement*).

(a) For $n = 3$, list all the derangements and find p_3. (3 points)

(b) For distinct guests $i_1, \dots, i_k$, explain why $\mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k}) = (n-k)!/n!$. Then use inclusion–exclusion to show that (5 points)

$$p_n = \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

(c) Evaluate p_4 as a fraction in lowest terms. (2 points)

(d) What is $\lim_{n\to\infty} p_n$? The error made by truncating an alternating series with decreasing terms is at most the first omitted term. Use this to find the smallest n for which this estimate guarantees that p_n is within $1/1000$ of its limit. (4 points)

(e) For $n = 4$, find the probability that nobody receives their own hat, given that guest 1 does not. (2 points)

4. Was the clerk careless?

10 points · about 10 minutes

A hat-check clerk is *careful* with probability 2/3 and *careless* with probability 1/3. A careful clerk returns every hat to its owner. A careless clerk returns the hats in a uniformly random order, as in Problem 3. Tonight there are $n = 3$ guests. Let L mean "the clerk is careless," let A_1 mean "guest 1 receives their own hat," and let C mean "all three guests receive their own hats."

(a) Find $\mathbb{P}(C \mid L)$ and $\mathbb{P}(C \mid L^c)$, and use the law of total probability to find $\mathbb{P}(C)$. (3 points)

(b) All three guests receive their own hats. Use Bayes' Rule to find the probability that the clerk was careless. (3 points)

(c) Suppose instead you learn only that guest 1 received their own hat. Find $\mathbb{P}(L \mid A_1)$. (2 points)

(d) Redo part (b) for a general number n of guests, and say what happens to $\mathbb{P}(L \mid C)$ as $n \to \infty$. Interpret the limit in one sentence. (2 points)